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Chapter 4

Product and Service Design

Operations Management

FOURTEENTH EDITION

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Chapter 4: Learning Objectives

You should be able to:

- Explain the strategic importance of product and service design
- Describe what product and service design does
- Name the key questions of product and service design
- Identify some reasons for design or redesign
- List some of the main sources of design ideas
- Explain the purpose and goal of life-cycle assessment
- Explain the phrase “the 3 Rs”
- Briefly describe the phases in product design and development
- Discuss several key issues in product or service design

Strategic Product and Service Design

- The essence of an organization is the *products* and *services* it offers, every aspect of the organization is structured around them.
- Well-designed products or services are more likely to realize their goals.
- Product and service *design* or *redesign* should be closely tied to an *organization's strategy*.

What Does Product & Service Design Do?

- Key requirements in the effective design of products and services include:

Translate **customer wants and needs** into product and service requirements

Develop new products and services

Refine existing products and services

Formulate **quality goals**

Formulate **cost targets**

Construct and test **prototypes**

Document **specifications**

Translate product and service specifications into **process specifications**

Involve inter-functional **collaboration**

Key Questions for design products and services

What is the **potential size** of the market for the product?

Will the demand for the product be **long-term** or **short-term**?

Does the organization have the necessary **knowledge, skills, equipment**, etc. to implement a potential product or service design?

Is it possible that all or part of the work could **be outsourced**?

What **level of product or service quality** is appropriate?

Does it make sense from an **economic standpoint**?

Reasons to Design or Re-Design

The driving forces for product and service *design* or *redesign* are market ***opportunities*** or ***threats***:

Economic

Social and demographic

Political and legal

Competitive

Cost or availability

Technological

Idea Generation

- Ideas for new or redesigned products or services can come from a variety of sources, including:

Customers

Supply
chain

Employees

Competitors

Research

Idea Generation: Supply-Chain Based

Ideas can come from anywhere in the *supply chain*:

- Customers
- Suppliers
- Distributors
- Employees
- Maintenance and repair personnel

Idea Generation: Competitor Based

Ideas can come from applying **Reverse engineering**:

- *Dismantling and inspecting* a competitor's product to discover product improvements.

Idea Generation: Research Based

Research and development (R&D)

- Organized efforts to increase scientific knowledge or innovation of product and service.
- **Basic research**
 - Has the objective of *advancing the knowledge about a subject* without any near-term expectation of commercial applications.
- **Applied research**
 - Has the objective of *achieving commercial applications*.
- **Development**
 - *Converts the results into useful commercial applications* through of applied research.

Legal and Ethical Considerations

- Product or service designers take into account:
 - Government *regulations*
 - Safety *standards*
 - Potential to harm the *environment*
 - Cut *costs*
 - Speed up the *process*
 - Adding new *features*
 - *Cultural* differences
- In general, these considerations are *necessary* and *mandatory*.

Legal Considerations

- **Product liability**

- The responsibility a manufacturer has for any *injuries* or *damages* caused by a faulty product.
- Some of the concomitant costs:
 - Litigation costs
 - Legal and insurance costs
 - Settlement costs
 - Costly product recalls
 - Reputation effects
- It's very important when designing products or service, *safety* and *high-quality* should be considered carefully.

Ethical Considerations

- **Designers are often under pressure to**
 - Speed up the design process
 - Cut costs
- **These pressures force trade-off decisions**
 - Product features V.S product design speed
 - Product quality V.S development costs

Sustainability of Product / Service

Sustainability issues

- Using resources in ways that do *not harm ecological systems* that *support human existence*.

Key aspects of designing for sustainability:

Life cycle
analysis
(LCA)

End-of-life
programs
(EOL)

The
3Rs

Life Cycle Analysis (LCA)

- The assessment of the **environmental impact of a product or service throughout its useful life.**



Life Cycle Analysis (LCA)

- LCA is to choose products and services that **have the minimum environmental impact while still considering important economic factors.**
- Focuses on such factors as:

Global
warming

Smog
formation

Oxygen
depletion

Solid waste
generation

End-of-Life (EOL) Programs

- Deal with products that have reached the end of their useful lives.
- These products may include:
 - Consumer products
 - Business equipment
- Improper disposal of waste products may cause *air* and *water* pollution.

The 3Rs

- Designers often reflect on three particular aspects of potential *cost savings* and *reducing environmental* impact.
 - **Reduce** costs and materials used
 - **Reuse** or **Refurbishing** parts of returned products
 - **Recycle** parts of unusable products



Reduce: Value analysis

- **Value analysis** is involves the examination of the function of parts and materials in an effort **to reduce the cost** and/or **to improve the *performance*** of a product
- Common questions used in value analysis:
 - Is the item *necessary*; does it *have value*; could it *be eliminated*?
 - Are there *alternative sources* for the item?
 - Could another material, part, or service be used *instead*?
 - Can two or more parts be *combined*?
 - Can specifications be *less stringent* to save time or money?
 - Do suppliers/providers have *suggestions for improvements*?
 - Can packaging *be improved or made less costly*?

Reuse: Remanufacturing

- **Remanufacturing** involves refurbishing products by replacing *broken, defective, or worn-out* components.
- Reasons to remanufacture:
 - Remanufactured products can be sold for about 50% of the cost of a new product.
 - In the global market, lawmakers are increasingly requiring manufacturers to take back used products.
- **Design for disassembly (DFD)**
 - Product design that takes into account *the ability to easily taken apart*.

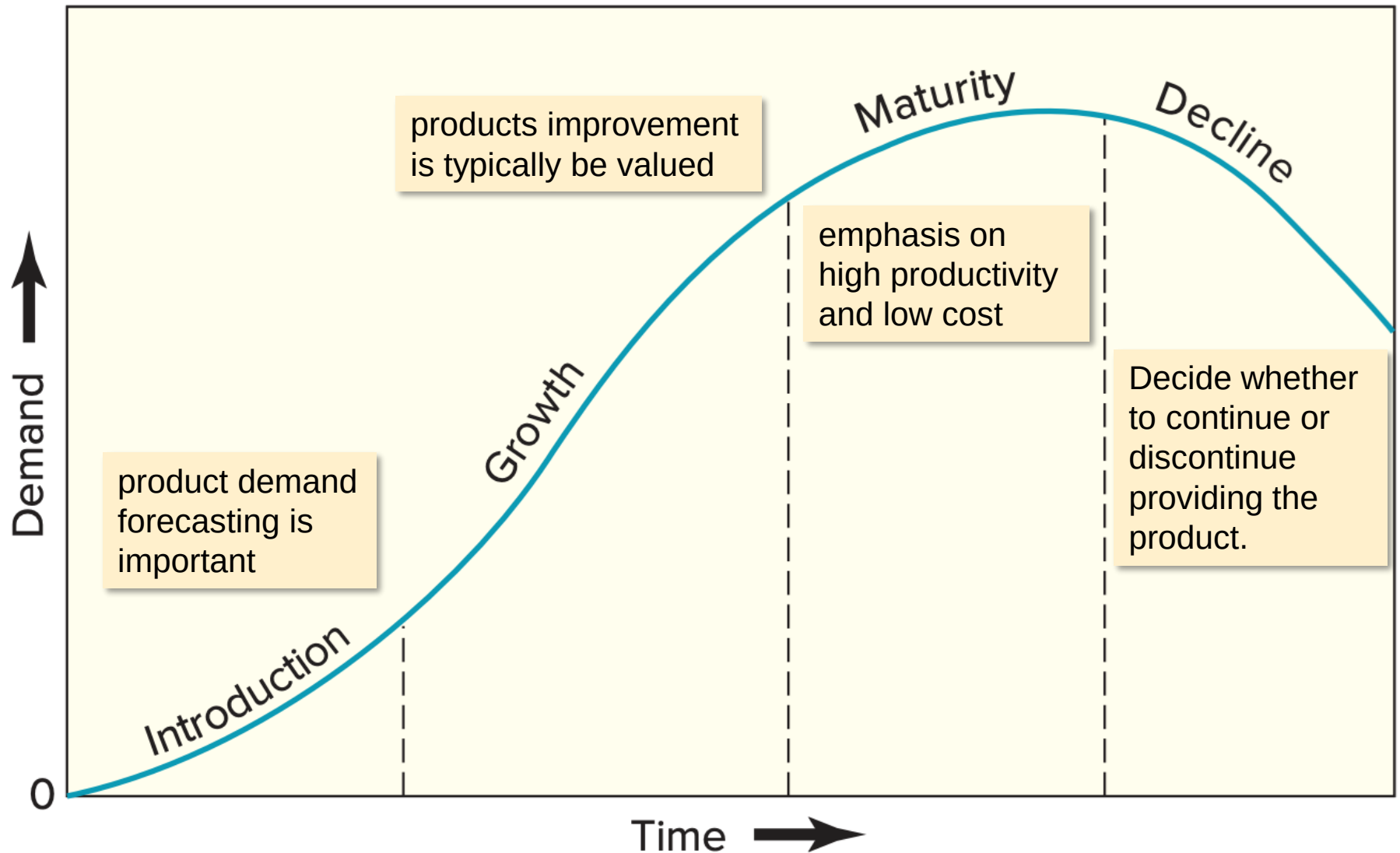
Recycle

- **Recycling** applies to both manufactured parts and materials used during production.
- Why recycle?
 - Cost savings
 - Environmental concerns
 - government regulations
- **Design for recycling (DFR)**
 - Product design that takes into account the ability to disassemble a used product to recover the recyclable parts.

Stages of Product Life Cycle (PLC)

- Products and services go through a series of stages over their useful life.
- **Product Life Cycle (PLC)**
 - *Demand typically varies by phase*
 - Different phases call for different *strategies*
 - *Forecasts of demand and cash flow are key inputs for strategy In each phase*

Stages of Product Life Cycle



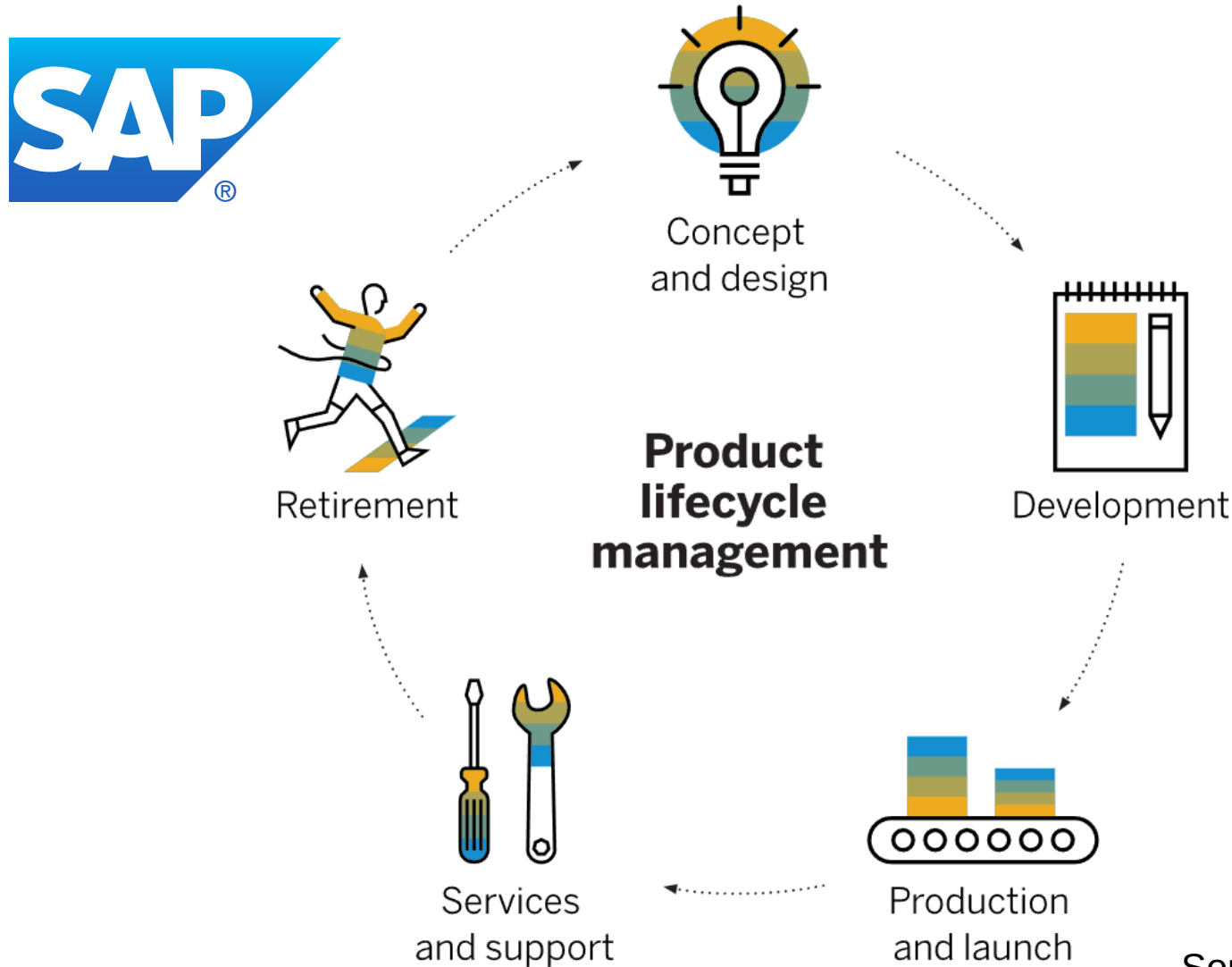
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Product Life Cycle Management (PLM)

- **PLM** is a systematic approach to managing the series of changes a product goes through in its lifecycle.
- The goal of PLM is to *eliminate waste* and *improve efficiency*.
- Three key phases of PLM application:



Product Life Cycle Management (PLM)



Source: [www. SAP.com](http://www.SAP.com)

Standardization

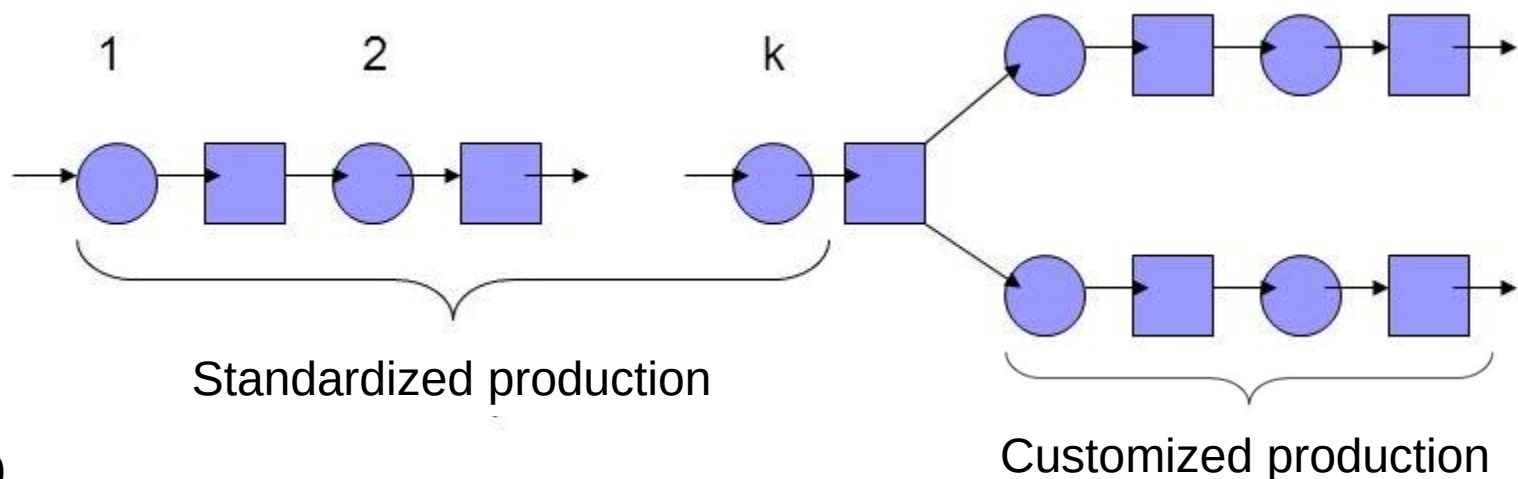
- **Standardization** is identical products or services that are produced in large volume or provided the same service.
- *Advantages*
 - permit products to be built according to forecasted demands
 - facilitate lower costs and increase productivity
 - make scheduling of work more routine
 - make replacement or repair easily
- *Disadvantages*
 - limits product variety
 - designs may be prematurely frozen
 - high cost of design changes

Mass Customization

- **Mass Customization** is a strategy of producing basically standardized goods or services, but **incorporating some degree of customization in the final product or service.**
- Facilitating techniques for mass customization
 - *Delayed differentiation*
 - *Modular design*

Delayed Differentiation

- **Delayed differentiation** is a process of **producing a product or service but not quite completing production until customer preferences are known.**
- It is a *postponement tactic* that instead of guessing what a customer might want, wait for the customer to reveal their preferences and then complete the final production.



Modular Design

- **Modular design** is a form of standardization in which component parts are grouped into modules that are easily replaced or interchanged.
 - Advantages
 - Easier diagnosis and remedy of failures
 - Easier repair and replacement
 - Simplification of manufacturing and assembly
 - Training costs are relatively low
 - Disadvantages
 - Limited number of possible product configurations
 - Limited ability to repair a faulty module



Robust Design

- **Robust design** is the process of trying to reduce variations and to maintain consistency of standards in the finished products.
- **Taguchi's Approach**
 - Proposed by Dr. Genichi Taguchi
 - Adopt parameter design

Taguchi's Approach

- Step 1: Conduct **Full Factorial Experiments**

Test	Crew Size	Capacity (hrs)	Output (units)
1	5	30	40
2	5	35	63
3	5	40	34
4	8	30	50
5	8	35	55
6	8	40	68
7	10	30	72
8	10	35	83
9	10	40	75

Taguchi's Approach

- Step 2: Perform **Analysis of Means**

Level	Test	Crew Size	Output (units)
LV1	1	5	40
	2	5	63
	3	5	34
LV2	4	8	50
	5	8	55
	6	8	68
LV3	7	10	72
	8	10	83
	9	10	75

Crew-based:

- $LV1_{Cr1} = (40 + 63 + 34) / 3 = 45.67$
- $LV2_{Cr2} = (50 + 55 + 68) / 3 = 57.67$
- $LV3_{Cr3} = (72 + 83 + 75) / 3 = 76.67$

Taguchi's Approach

- Step 2: Perform **Analysis of Means**

Level	Test	Capacity (hrs)	Output (units)
LV1	1	30	40
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	9	40	75

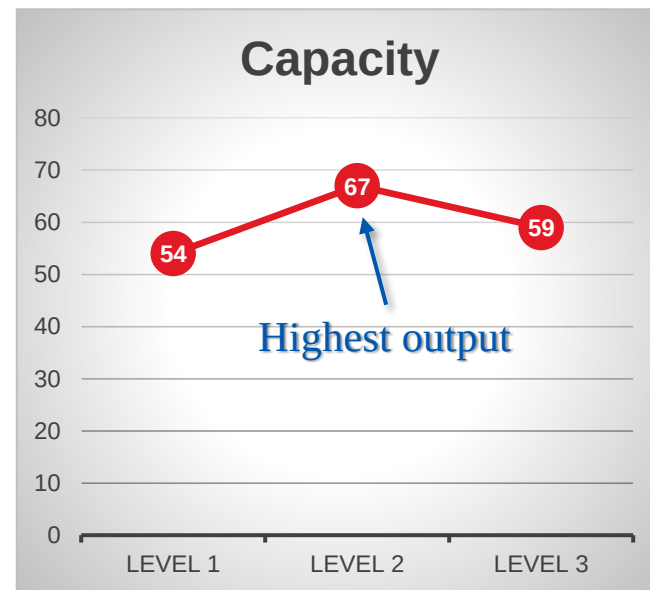
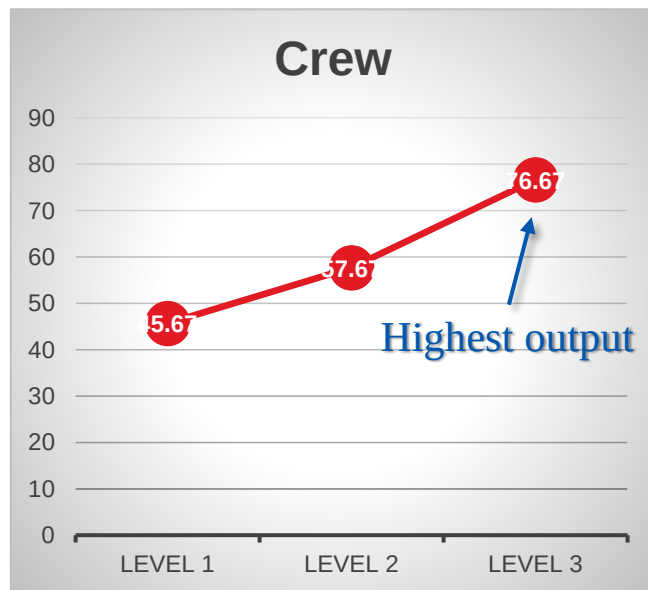
Capacity-based:

- $LV1_{Ca1} = (40 + 50 + 72) / 3 = 54.00$
- $LV2_{Ca2} = (63 + 55 + 83) / 3 = 67.00$
- $LV3_{Ca3} = (34 + 68 + 75) / 3 = 59.00$

Taguchi's Approach

- Step 3: Building a **Response Table** and **Effect Plot**

Factor	Level 1		Level 2		Level 3	
	Input	Output	Input	Output	Input	Output
Crew size	5	45.67	8	57.67	10	76.67
Capacity	30	54.00	35	67.00	40	59.00



Quality Function Deployment (QFD)

- **Quality Function Deployment (QFD)** is an approach that integrates the “*voice of the customer (VOC)*” into both product and service development.
 - To ensure that customer requirements are considered into every aspect of the process.
 - To listen and to understanding the customer is the central feature.

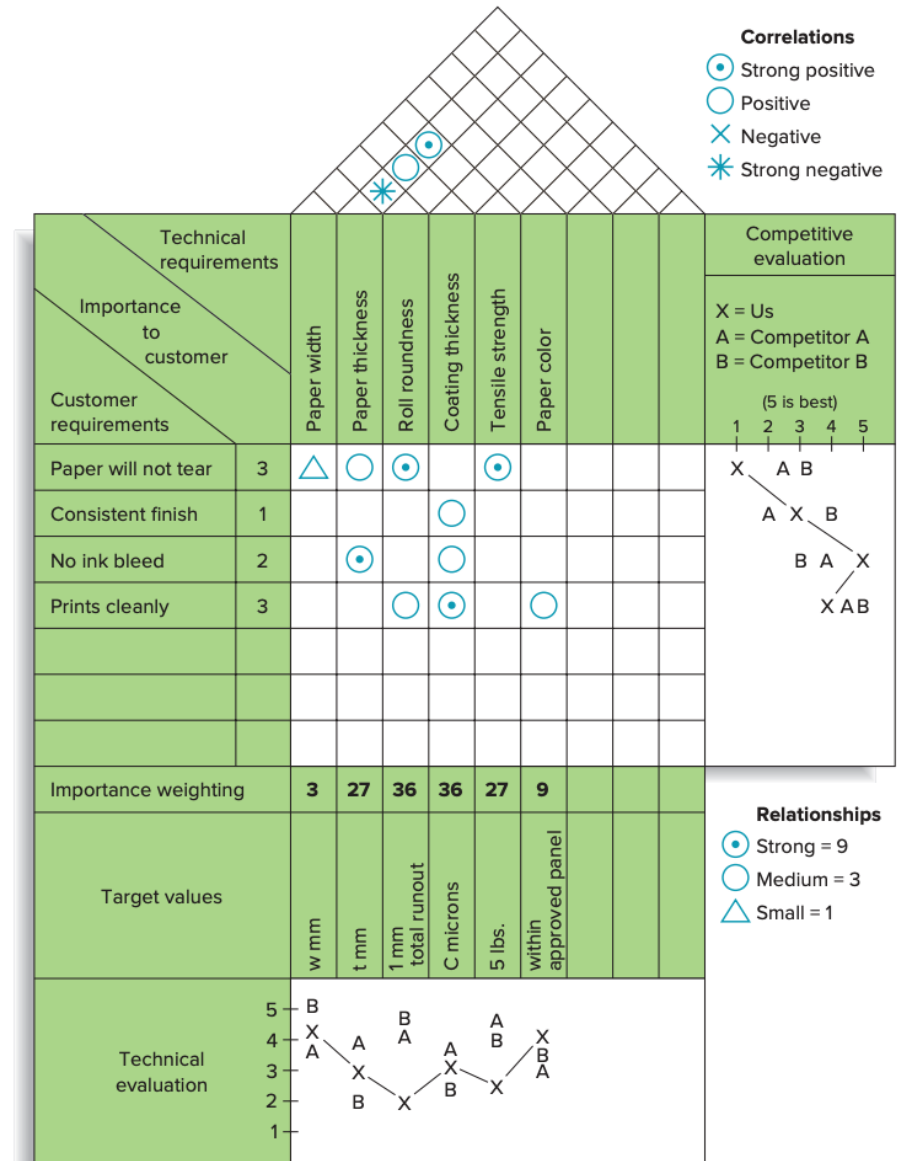
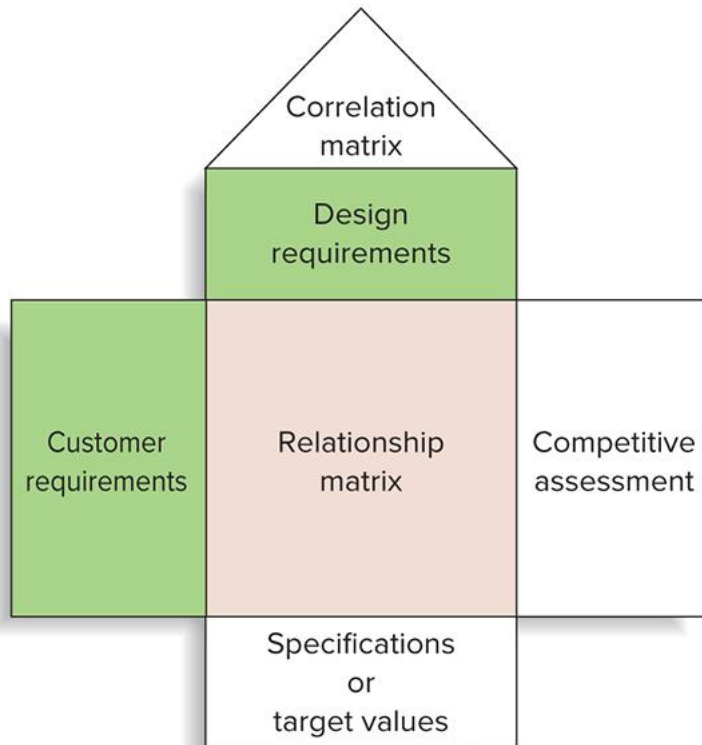
Quality Function Deployment (QFD)

- **House of Quality**
 - Proposed by Mitsubishi Heavy Industry
 - The basic design tool of QFD
- **Kano Model**
 - Proposed by professor Noriaki Kano
 - A theory on quality design of product and service

House of Quality

- The output is generally a matrix with customer desires on one dimension and correlated functional requirements on the other dimension.
 - Classifies customer desires and identifies the importance of those desires (*What's*).
 - Identifies product features or functions which may be relevant to those desires (*How's*).
- Combines correlations of customer needs and product functions and assigns objectives and priorities for the product requirements.

House of Quality



Kano Model

- **Basic quality**

- Refers to customer *basic requirements* that have only limited effect on customer satisfaction if present, but lead to dissatisfaction if absent.

- **Performance quality**

- Refers to customer satisfaction or dissatisfaction based on whether the *product's functionality meet their requirements*.

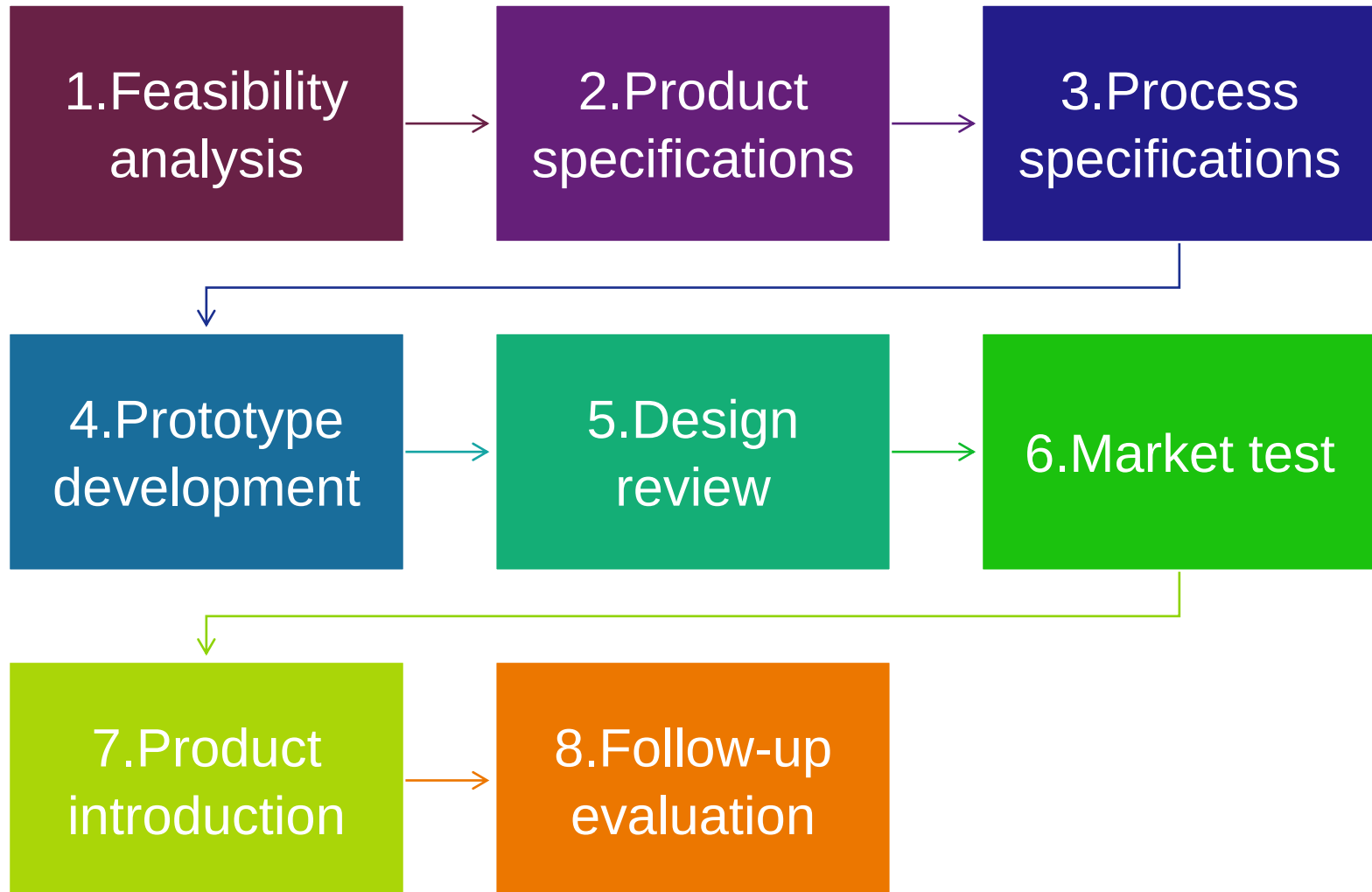
- **Excitement quality**

- Refers to a *feature or attribute that was unexpected* by the customer and causes excitement.

The Kano Model – As Time Passes



Phases in Product Design and Development



Phases in Product Design and Development

Phase 1: Feasibility analysis

- Feasibility analysis contains **market analysis**, **economic analysis**, and **technical analysis**.
- It is necessary to answer the question:
 - Does it fit with the organization's mission?

Phase 2: Product specifications

- Product specifications involves **detailed descriptions of what is needed to meet customer wants**.
- It requires collaboration between legal, marketing, and operations.

Phases in Product Design and Development

Phase 3: Process specifications

- Process specifications involves to **formulate process for producing a product or designing a service.**
- It must be weighed in terms of cost, resources availability, profit potential, quality.
- Requires collaboration between accounting and operations.

Phases in Product Design and Development

Phase 4: Prototype development

- With product and process specifications complete, one (or a few) **prototypes of products are made to see if there are any problems with the product or process specifications.**

Phase 5: Design review

- At this phase, **any necessary changes are made or the project is abandoned.**
- Marketing, finance, engineering, design, and operations collaborate to determine whether to proceed or abandon.

Phases in Product Design and Development

Phase 6: Market test

- Market test is **used to determine the extent of consumer acceptance**, and handled by marketing.
- If unsuccessful, the product returns to the design review phase.

Phase 7: Product introduction

- Once passed the phase of market test, **new product will be ready for release, it handled by marketing.**

Phases in Product Design and Development

Phase 8: Follow-up evaluation

- Based on users' feedback, products or services may require ongoing updates, revisions, or redesigns, until the end of the product life cycle.

Differences between Service and Product Design

- Services are *intangible*; products are generally *tangible*
- Services are created and delivered *at the same time*
- Services cannot be *inventoried*
- Services are *highly visible* to consumers
- Some services have *low barriers to entry and exit*
- *Location is often important* to service design
- Service is more closely *connected with customers*

Operations Strategy

- **Effective product and service design can help the organization achieve competitive advantage:**
 - Packaging products and ancillary services to increase sales.
 - Implementing tactics that will achieve the benefits of high volume while satisfying customer needs for variety.
 - Continually monitoring products and services for small improvement opportunities.
 - Reducing the time it takes to get a new or redesigned product or service to the market.

Summary

- Product and service design is a key factor in satisfying the customer. To be successful in product and service design, organizations must be continually aware of what customers want, what the competition is doing, what government regulations are, and what new technologies are available.
- The design process involves motivation, ideas for improvement, organizational capabilities, and forecasting. In addition to product life cycles, legal, environmental, and ethical considerations influence design choices.

Summary

- What degree of standardization designers should incorporate into designs is also an important consideration.
- A key objective for designers is to achieve a product or service design that will meet or exceed customer expectations, within cost or budget and taking into account the capabilities of operations.
- Although product design and service design are similar in some respects, a number of key differences exist between products and services that influence the way they are designed.

Summary

- Successful design often incorporates many of these basic principles: Determine what customers want as a starting point; minimize the number of parts needed to manufacture an item or the number of steps to provide a service; simplify assembly or service, standardize as much as possible; and make the design robust.
- Trade-off decisions are common in design, and they involve such things as development time and cost, product or service cost, special features/performance, and product or service complexity.

Summary

- Research and development efforts can play a significant role in product and process innovations, although these are sometimes so costly that only large companies or governments can afford to underwrite them.
- Reliability of a product or service is often a key dimension in the eyes of the customer. Measuring and improving reliability are important aspects of product and service design, although other areas of the organization also have an influence on reliability.
- QFD is one approach for getting customer input for product or service design.



The End